

Word Embeddings for Code-Mixed Language Processing

Adithya Pratapa, Monojit Choudhury, Sunayana Sitaram
Microsoft Research, India
{t-pradi, monojitc, sunayana.sitaram}@microsoft.com

1. Introduction

- Word Embeddings** are dense representations for words capturing various syntactic and semantic properties
- they allow for generalizing the system on a much larger corpora than the annotated data available for the task
- Code-Mixing (CM)** refers to juxtaposition of linguistic units from two or more languages in a single utterance. Few Spanglish (Spanish-English) examples,
 - pero **I'm glad** que lo aceptó <but I'm glad you accepted it>
 - el trabaja más en **background** eh como investigador **approving the files** <he works more in background eh as a researcher approving the files>
 - pero **I thought** ella fue **like very good** <but I thought she was like very good>
- A lot of prior work in learning monolingual and cross-lingual word embeddings (Upadhyay et al. 2016, Ruder et al. 2017)



Can we use the existing bilingual/cross-lingual approaches for learning CM word embeddings?

2. Existing Bilingual Embedding Approaches

- Bilingual CCA (BiCCA)**
 - Bilingual evidence is incorporated into the word representations by performing canonical correlation analysis (CCA) on monolingual embeddings using a bilingual dictionary (Faruqui and Dyer, 2014)
$$\mathbf{P}_{L_1}, \mathbf{P}_{L_2} = CCA(\mathbf{W}'_{L_1}, \mathbf{W}'_{L_2})$$
$$\mathbf{W}^S_{L_1} = \mathbf{W}_{L_1} \mathbf{P}_{L_1}, \quad \mathbf{W}^S_{L_2} = \mathbf{W}_{L_2} \mathbf{P}_{L_2}$$
- Bilingual CVM (BiCVM)**
 - Parallel sentences from the two languages should have equivalent meanings $\rightarrow$ similar sentence representations (Hermann and Blunsom, 2014)
$$E_{bi}(s_1, s_2) = \|\mathbf{f}_{L_1}(s_1) - \mathbf{f}_{L_2}(s_2)\|^2$$
$$L(\mathbf{W}_{L_1}, \mathbf{W}_{L_2}) = \|\mathbf{W}_{L_1}\|^2 + \|\mathbf{W}_{L_2}\|^2 + \sum_{(s_1, s_2) \in A, k} \max(0, 1 + E_{bi}(s_1, s_2) - E_{bi}(s_1, s_n^k))$$
- Bilingual Skip-gram (BiSkip)**
 - Adapting Skip-gram to bilingual setting by learning to predict word contexts cross-lingually (Luong et al. 2015)
$$L_{12}(\mathbf{W}_{L_1}, \mathbf{W}_{L_2}) = - \sum_{(w_1, w_2) \in A, w_c \in context(w_2)} \log(P(w_c | w_1))$$

la modestia es una cualidad tan hermosa
modesty is such a beautiful quality

Some phrases are more acceptable (or frequent) for substitution than others,

modesty es una cualidad tan hermosa

modesty is such **una** beautiful quality
la modestia **is** una cualidad tan hermosa

Challenges for Code Mixed data

- The existing approaches are primarily designed for cross-lingual applications and there are emergent syntactic structures and cross-lingual semantic associations in CM text that don't appear in monolingual corpora
- There is a paucity of CM data in written text form to directly learn embeddings

3. Synthetic CM based Embeddings

- Synthetic CM data**
 - Real CM data is scarce in written text form
 - So, we generate CM data from monolingual parallel data by applying Equivalence Constraint theory (Pratapa et al. 2018)
 - They account for the syntactic constraints imposed on CM data
- Data and Sampling**
 - The synthetic data generation process utilizes parallel corpora, along with word-level alignments and English parses.
 - It constructs a word lattice of *grammatically* valid CM sentences
 - #{sentences} generated are in the order of exponential of #{monolingual pairs} and language usage statistics are different
 - Hence, Sampling, (k)
 - Random (**χ -gCM**)
 - SP (switch point) fraction based (**ρ -gCM**)
- Embedding training data**
 - 4.5M parallel sentences with word level alignments
 - Trained a skip-gram model with negative sampling on the sampled synthetic data (with k=2, around 8M synthetic + 9M monolingual)

4. Quantitative Evaluation

- Sentiment Analysis (semantic task)**
 - Data:** 2103 sentiment annotated tweets (650 positive, 529 negative and 924 neutral) (Vilares and Alonso, 2016)
 - Eso es lo q significa q un rayo puede caer, You never know lighting could strike, figurativamente es q lo improbable puede suceder* (Neutral)
<That's what it means that lightning can fall, You never know lighting could strike, it's figuratively that the unlikely can happen>
 - Como pasa el tiempo, y con el cosas en la vida k te cambian para siempre.. But hey I got a second chance! :) I cant believe its been a year* (Positive)
<As time passes, and with things in life, they change you forever ... But hey I got a second chance! :) I cant believe its been a year>
 - Model:** LSTM based sequence classifier
 - Evaluation on Monolingual tasks:** English (SemEval 2014 Subtask B) and Spanish (TASS 2016) to check the robustness of the embeddings
- POS tagging (syntactic task)**
 - Data:** Bangor Miami Corpus, only CM utterances with significant fraction of English and Spanish ($\geq 30\%$). 982 sentences with 7705 tokens
 - y supuestamente they are in the process de hacer ese cambio*
<and supposedly they are in the process of making that change>
y:spa:CONJ supuestamente:spa:ADV they:eng:PRON are:eng:V in:eng:PREP the:eng:DET process:eng:N de:spa:PREP hacer:spa:V ese:spa:ADJ cambio:spa:N
 - Model:** Bi-directional LSTM model (Rei and Yannakoudakis, 2016) with CRF as the output layer

5. Results

Embedding	Sentiment			POS	
	CM Overall	SemEval 2014	TASS 2016	CM Overall	SP
None	54.4 (1.3)	64.5 (0.6)	61.4 (1.0)	84.5 (0.3)	74.0 (0.7)
BiCCA	57.6 (3.0)	64.6 (1.0)	59.5 (1.8)	84.7 (0.8)	75.0 (1.8)
BiCVM	64.3 (1.3)	66.8 (1.0)	61.9 (1.0)	82.0 (0.5)	70.6 (1.7)
BiSkip	61.5 (1.7)	66.6 (0.9)	63.9 (1.2)	84.4 (0.7)	73.8 (0.9)
χ -gCM-Skip	62.0 (1.9)	67.4 (1.3)	63.2 (1.5)	84.8 (0.6)	74.0 (0.6)
ρ -gCM-Skip	64.6 (2.0)	67.7 (1.4)	63.8 (2.2)	84.9 (0.7)	75.3 (1.7)

Table 1: performance of pretrained embeddings on Sentiment (F1 score) and POS (accuracy) tasks. The reported values are mean and deviation (in parenthesis) over multiple runs

- ρ -gCM-Skip does consistently well across tasks, while BiCVM does well on Sentiment task and BiCCA performs well on POS tagging task
- In POS tagging task, prediction is challenging at SP (switch point)
- ρ -gCM can be a reasonably good proxy for real CM data

6. Open Problems

- Linguistic theory based synthetic data generation only handles the syntactic constraints of CM
- Other fronts to improve,
 - Lexical choice:** could be skewed towards one language given a CM context (Bali et al. 2014)
 - Semantic/Pragmatic constraints:** use of native language for expressing particular opinions (Rudra et al. 2016)
 - Sense distribution:** could be very different between monolingual and CM context
- Future directions,
 - Generating synthetic data (for training) that approximates lexical, semantic and syntactic constraints together
 - Building embedding techniques that can interpolate between real and synthetic CM data to account for all the constraints

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